Investigating accelerometer accuracy by integrating twice

November 9, 2026

1 Introduction

In Vex robotics, there are sensors for quickly measuring locations for accurate repeated robotic automationos. And the most useful one is an IMU. An inertial measurement unit (IMU) is a sensor that measures acceleration at a high rate. Whenever you need to know the precise location of something quickly, integrating accelerometer data twice to find displacement outpaces GPS solutions. They rely on the property that the area under a accelerometer graph can provide enough data used to compute the displacement of the object. In this investigation we will explore numerical and analytical integration of the accuracy of accelerometer data recorded on a phone accelerometer and comparing with video recorded displacement.

2 How to find displacement from acceleration graph

Recall the fact that velocity is the integral of acceleration and displacement is the integral of velocity. Integral is the area under a curve. For continuous functions, they are an approximateted by dividing the function into many "rectangles" like shown below where the height of a specific rectangle at a specific point of the function is the y value of the function

As the width of the rectangles becomes smaller and smaller, the accuracy increases. So in terms of collecting our data, having data with higher sample rate then it would become more accurate.

But instead of using rectangles to find the area under the curve, we can just use basic geometric equations like the equation for finding the area of a triangle and rectangle.

$$A_{triangle} = \frac{1}{2} b_t \cdot h_t \tag{1}$$

Figure 1: The formula finding the area of a triangle

$$A_{rectangle} = b_r \cdot h_r \tag{2}$$

Figure 2: Formula for the area of a rectangle

the subscripts r and t indicate which base we are refering to (for instance it is the triangles rather than the rectangle's)

Lets say we have two data points, the second data point's y-value could either be higher, lower, or equal to the y-value of the first point (shown below). And the numerical integral is essentially the area under between these two points, which can be calculated with the previous geometric equations illustrated below labeled with the equations

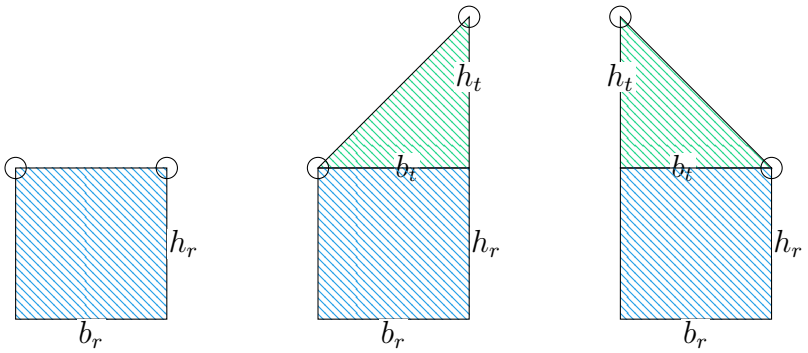


Figure 3: The three conditions of integration

But of course our data isn't just two points, but for the first two points, we find the area, and then the integral function's will use that area as the y-value for the new point. And the point's x-value should be in the end of the two points that we calculated the area with because its refering to the area based on what has happened before. And if we have another point, the integral function's corresponding new point's y-value will be the previous point's y-value plus the area between the new point and the previous point, and once again the x-value will be in between the two points in which the area is calculated from. So this is the data we collected from swinging the phone right to left 5 times. Positive means to the e Shown in Figure 4

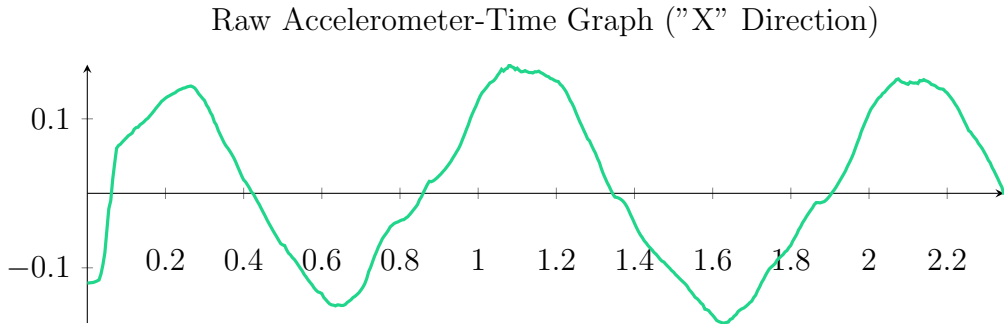


Figure 4: Raw accelerometer data on the X axis

So now have the tools to numerically integrate a piece of data, giving us a velocity-time graph. Shown below in Figure 5

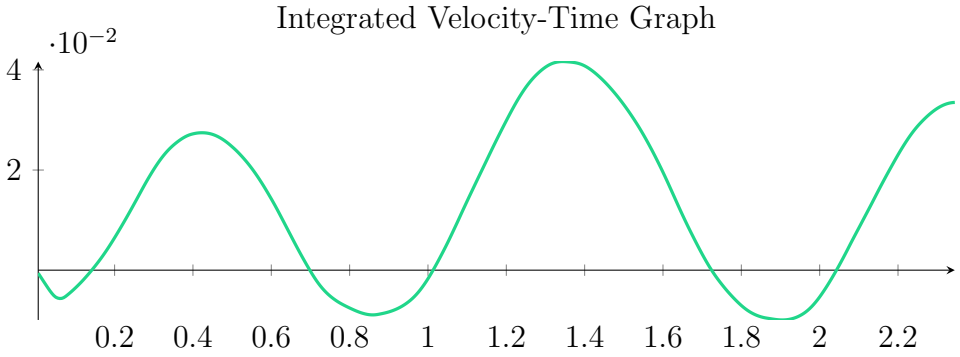


Figure 5: Integrated accelerometer data (velocity)

In theory, if we integrate again, we can finally get a displacement-time graph. But the velocity graph feels a little off because the area is mostly on the upper side of the x-axis, meaning if we were to integrate it, the displacement would start going towards the right rather than what actually happened: going right and left 5 times and then stopping.

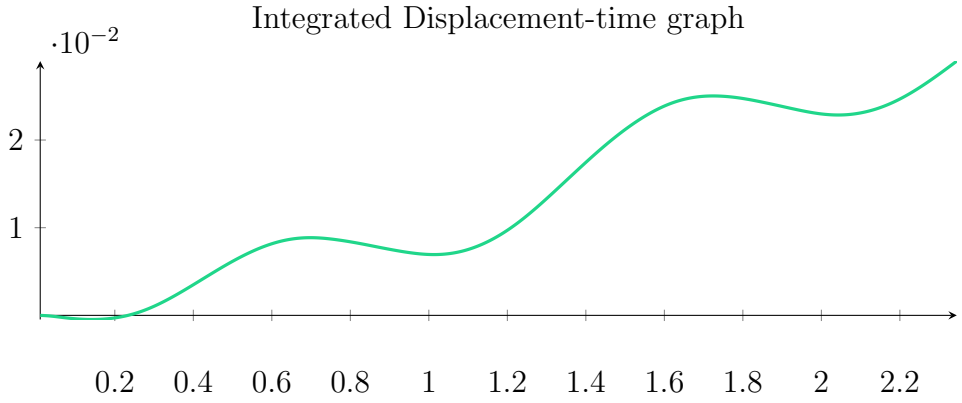


Figure 6: Integrated accelerometer data (Displacement)

One way I thought of to correct this is to center the accelerometer-time graph before integrating it into velocity. As you can see below

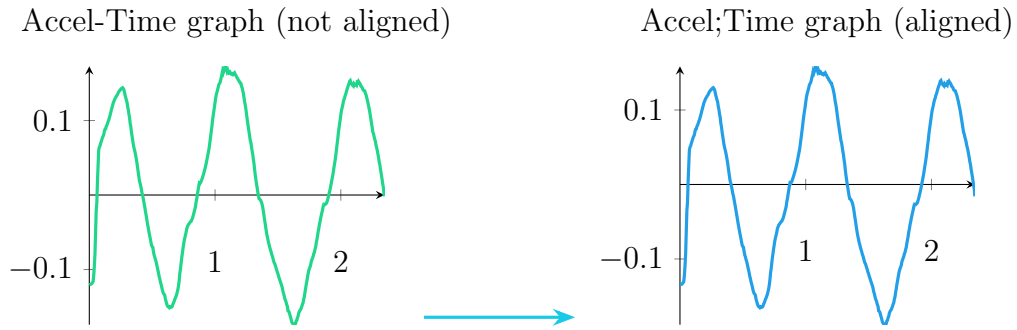


Figure 7: Raw accelerometer data on the X axis

And now we try integrating again:

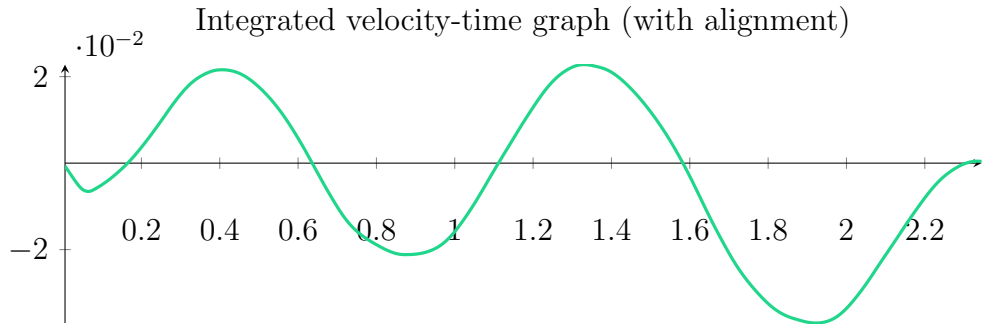


Figure 8: Integrated velocity (aligned)

Ok, much better. We should probably center it again before integrating it once more to get the displacement-time graph.

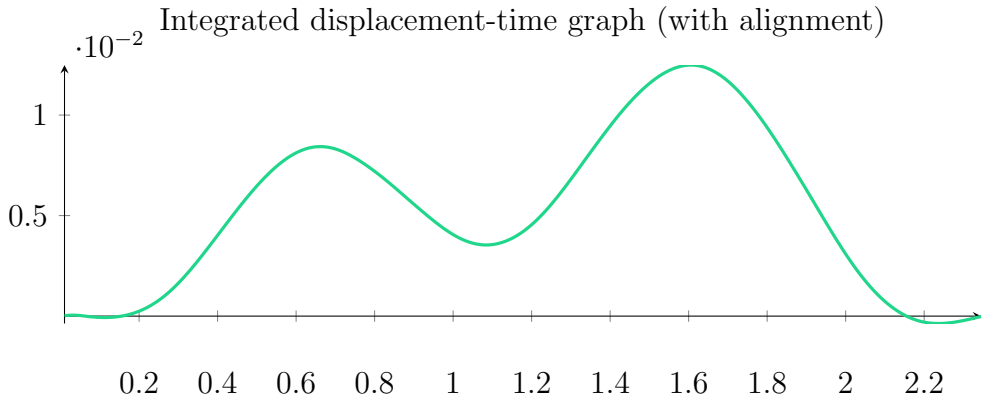


Figure 9: Integrated displacement (aligned)

Okay, this graph doesn't really seem like its' mean is zero even when the velocity-time graph has centered the mean to be zero. But there is no mistake, we look at the first up and down of the integrated velocity function (from $0 < x < 0.6$), it is indeed positive, which makes us reach a maxima of $(0.6, 1)$ in the displacement-time graph. In the integrated velocity function, the next part's area is negative (from $0.6 < x < 1.1$), and it subtracts the integrated displacement graph back to being close to zero, but doesn't ever go below it reaching a minimum close to but not reaching zero, and since it repeats going up and down and up, again, and we know that the minima is not lower than zero, it effectively makes the mean, not zero. But what if I

wanted the displacement's mean to actually be zero? Well, what makes the mean non zero in the first place? If we look the velocity-time graph as a sine function, then we will notice it is made up of four parts:

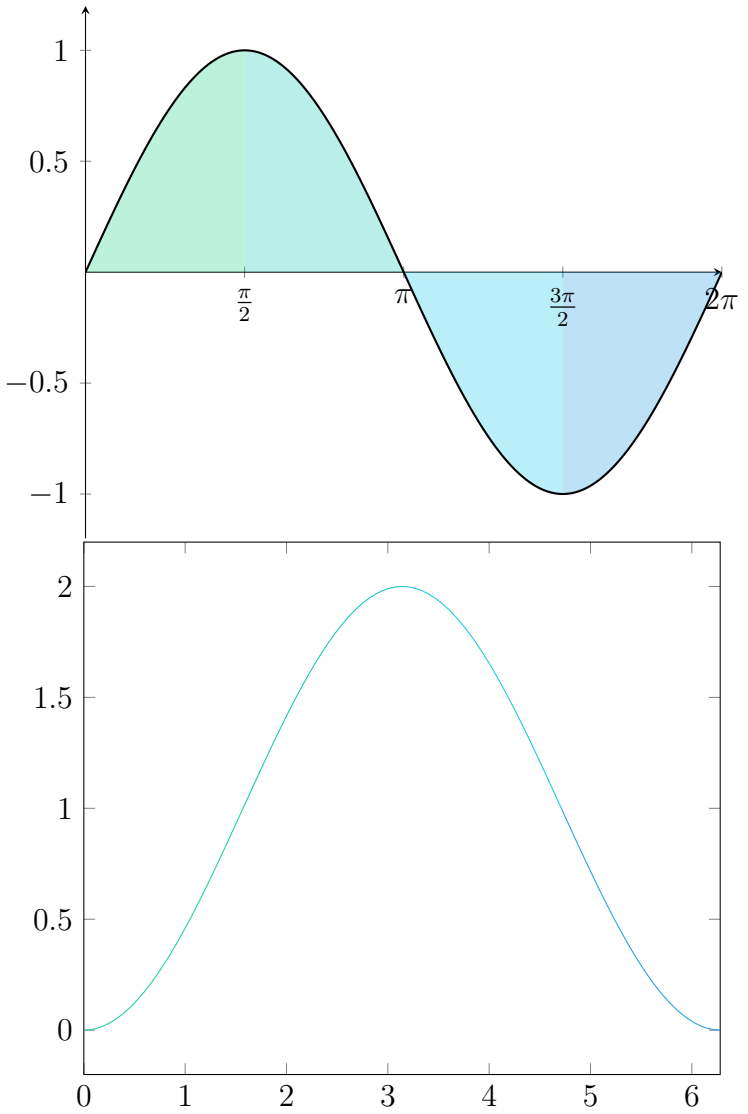


Figure 10: Four parts of the sine function

Since we start the function at initial displacement = 0, the amount of area added by the first two parts (green and teal parts) determines the maximum

of the graph; because we get subtracted by the same amount by the last two parts (cyan and blue), we go back to where we started (initial displacement = 0), which is why the initial displacement if now our minimum. But if we shift the phase of the sine function by $pi/2$ then we get a graph like this:

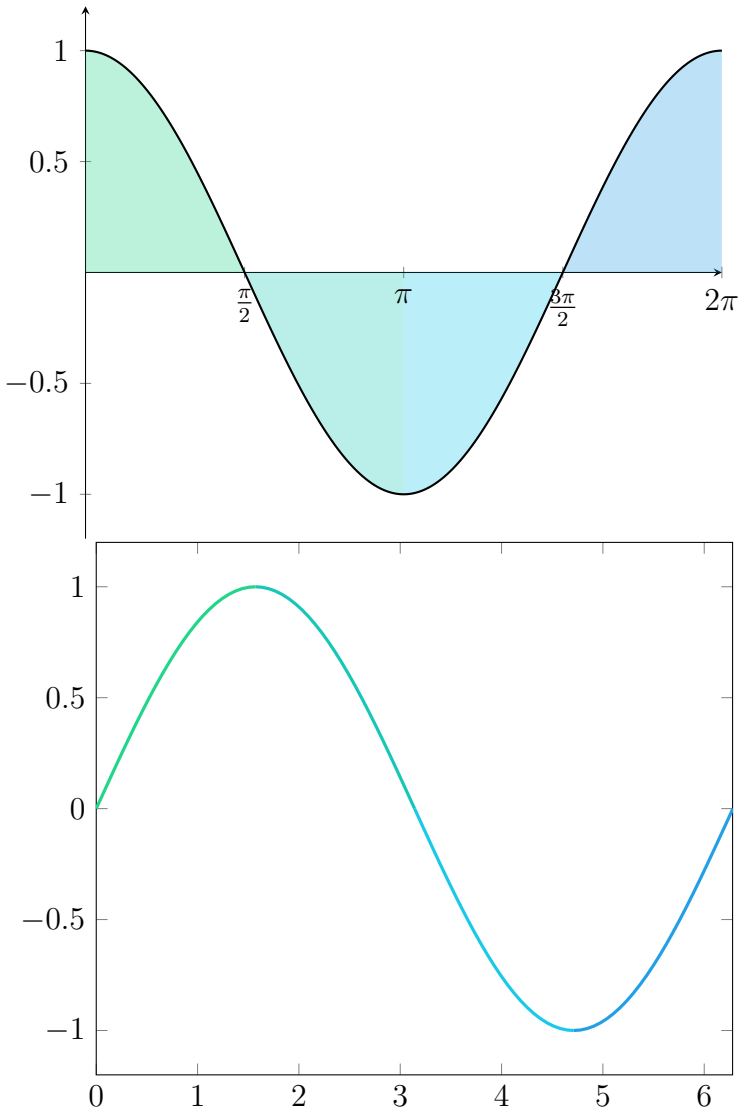


Figure 11: Four parts of the $\sin(x + \frac{pi}{2}$ function

Here the first two parts' area will be zero (since they cancel each other

out), and so does the second one, so in this case, instead of our initial displacement setting the mean, the initial displacement now sets the mean.

Lets compare this data with the actual displacement-time data I recorded on video.

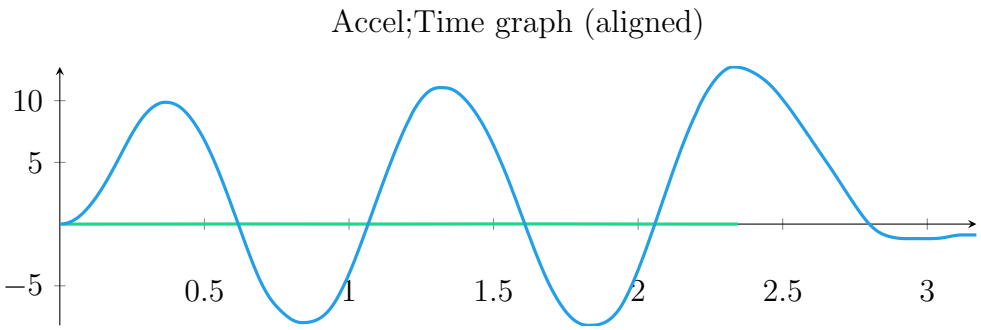


Figure 12: Comparison of integrated displacement against actual displacement

The y-values seem very tiny for the integrated displacement as compared to the recorded values for some reason, this is likely an error on the integration algorithm I made.

3 Modeling

I decided to compare integrating numerically against analytically; I want to know if there is a difference if I first modeled it with a sine equation and then integrated the equation algebraicly rather than numerically using geometry like I did before.

Desmos has a tool for fitting data into many types of equations. In particular, we are looking at a sinulsoidal regression

$$y = a\sin(bx + c) + d \tag{3}$$

Plugging into desmos for sinusoidal regression we get:

$$y = 0.14272 \cdot \sin\left(6.6486x + 0.216978\right) - 0.00389886 \tag{4}$$

seen in figure 13.

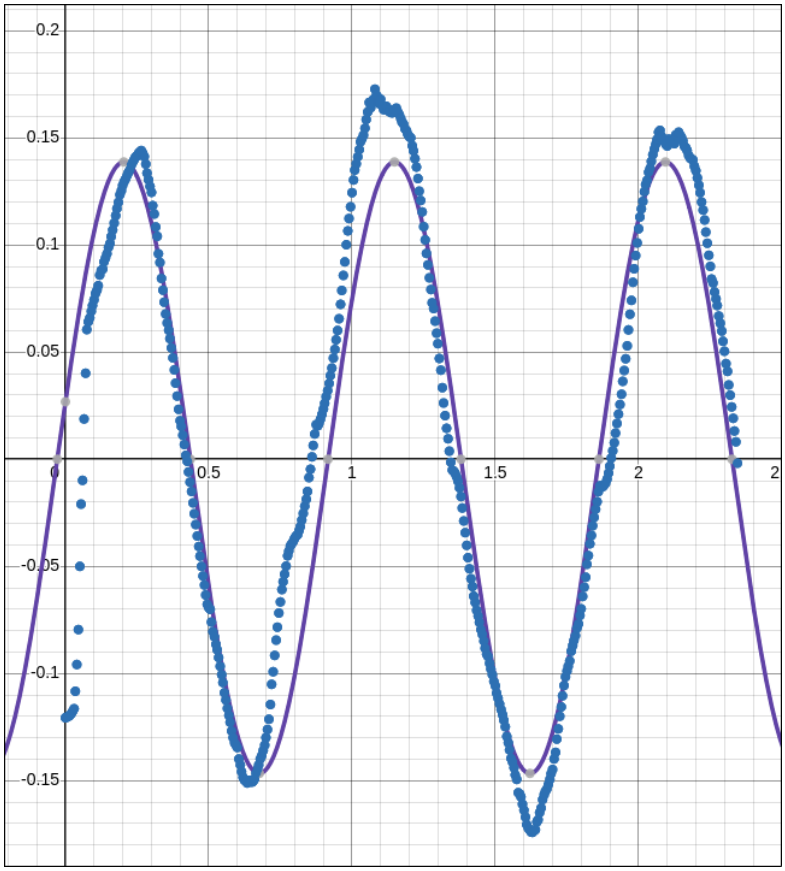


Figure 13: graph of our accelerometer data and the fitted curve

Intuitively, the last d term represents sensor drift, if we integrated $sin(x)+1$ to find velocity $v(t)$:

$$v(t) = \int sin(x) + 1 \, \mathrm{d}x \tag{5}$$

$$= -cos(x) + x + c \tag{6}$$

then, it's graph

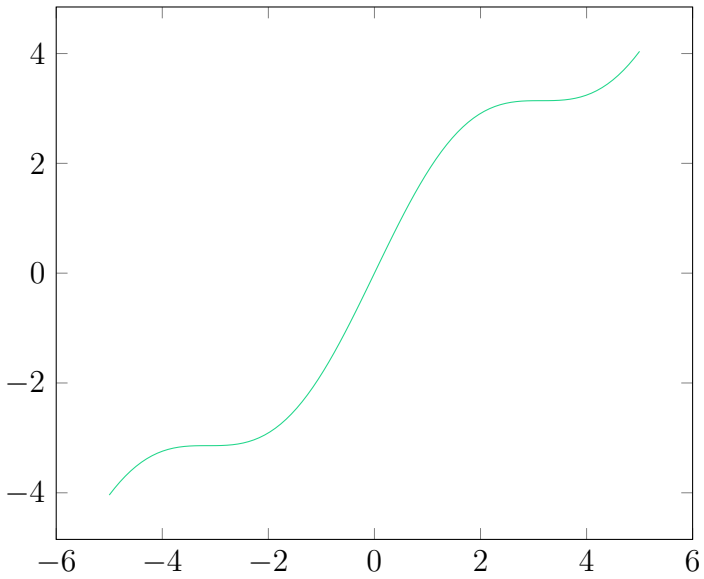


Figure 14: graph of $\sin(x) + x$

would just go up indefinitely, which wouldn't make sense especially when I moved my the phone back and forth and ended at rest because if the velocity never comes back down to zero, it would never come back to rest. If we just remove the d term in $a\sin(bx + c) + d$ then, its integration won't have any drift. Removing the d term in this sinusoidal regression before integration is akin to centering the accelerometer data before integration in the numerical method.

4 Integrating to velocity

We will need to integrate once to get velocity:

$$v(t) = \int (a \cdot \sin(bt + c) - d)\,\mathrm{d}t \tag{7}$$

where a, b, c, and d are constants. Rearranged, we get

$$v(t) = a\int \sin(bt + c)\,\mathrm{d}t - \int d\,\mathrm{d}t \tag{8}$$

Using u-substitution

$$u = bt + c \tag{9}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = b \tag{10}$$

$$\mathrm{d}t = \frac{\mathrm{d}u}{b} \tag{11}$$

plugging in u and dt:

$$v(t) = a \int \sin(bt + c)\,\mathrm{d}t - \int d\,\mathrm{d}t \tag{12}$$

$$= a \int \sin(u) \frac{1}{b}\,\mathrm{d}u - \int d\,\mathrm{d}t \tag{13}$$

$$= \frac{a}{b} \int \sin(u)\,\mathrm{d}u - \int d\,\mathrm{d}t \tag{14}$$

$$= -\frac{a}{b}\cos(u) - \int d\,\mathrm{d}t \tag{15}$$

$$= -\frac{a}{b}\cos(bt + c) - dt + C_1 \tag{16}$$

we get our velocity time equation

$$v(t) = -\frac{a}{b}\cos(bt + c) - dt + C_1 \tag{17}$$

In kinematics, the constant C_1 is the initial velocity, and since we know that our initial velocity is 0, we can ignore C_1 and set it to zero...is what I would say if I didn't pay enough attention. Because if we find the constant C_1 of our modeled function when initial velocity $v(0)$ is actually $= 0$

$$v(0) = -\frac{a}{b}\cos((0)\,b + c) + C_1 = 0 \tag{18}$$

$$\frac{a}{b}\cos(c) = C_1 = 0.0209628448456 \tag{19}$$

$$\tag{20}$$

It is not actually zero! That's because since our initial acceleration (defined by $a\sin(bt+c)+d$) depends on both c and d to determine initial acceleration, and even when $d = 0$, $c \neq 0$ which means our initial acceleration isn't actually zero, but why would that mean our initial velocity C_1 isn't zero then?

5 A conversation with C_1

Okay. The problem is that if we have a C_1 that is not 0, then this would mean that our displacement graph will drift because it means that the area of the graph is either more positive or more negative, which adds up over time. How do we make sense of C_1 then? And if I remember that the phone ended in the same place it started, then why would C_1 be nonzero? There's the obvious answer of uncertainty at play, but what does $C_1 \neq 0$ imply about our mathematical model?

Earlier, we solved

$$v(0) = -\frac{a}{b}\cos((0)\,b + c) + C_1 = 0 \tag{21}$$

$$\frac{a}{b}\cos(c) = C_1 = 0.0209628448456 \tag{22}$$

$$\tag{23}$$

By adding C_1 here it assumes that the antiderivative $F(x) = -\frac{a}{b}\cos((0)\,b + c)$ (without the C_1) is unable to be zero, meaning that we assume integrating the acceleration model wouldn't result in the initial velocity being zero $\int a(t)\Big(t = 0\Big) = v(0) = 0$ and that we need an additional term C_1 to vertical shift $v(t) = F(t)$ by C_1 such that the initial velocity of the velocity function is zero. So the mathematical reason why $C_0 \neq 0$ is because $F(0) \neq 0$ meaning $F(x) = -\frac{a}{b}\cos((0)\,b + c) \neq 0$. If we do want to make $C_1 = 0$, this means we need $-\frac{a}{b}\cos((0)\,b + c) = 0$. And since we know that a and b are responsible for the horizontal/vertical dilation, we know it cannot be zero, so we ignore that for now. Leaving us with

$$\cos((0)\,b + c) = 0\cos(c) = 0c = \frac{\pi}{2} \tag{24}$$

which means in order for $C_1 = 0$ we need $c = \frac{\pi}{2}$. Which is still fine, because acceleration could start a a non zero value, it just means that it experienced a force. Which makes sense because for our velocity function

But it does raise an implication. It implies that the phase shift c is able to impact the drift C_1 of the integrated velocity function if the velocity function has a condition $v(0) = 0$. It makes sense because the antiderivative of sin(x+c) is -cos(x+c), so at the end, we are just saying that by changing (the phase shift of the original function), we are going to start at different initial velocities.

For instance, if you didn't accelerate in the beginning, represented by $\sin(x)$, but you still start at zero velocity, then the integration from 0 to π will be purely positive and the minima will be 0 rather than -1. Illustrated as the green part below:

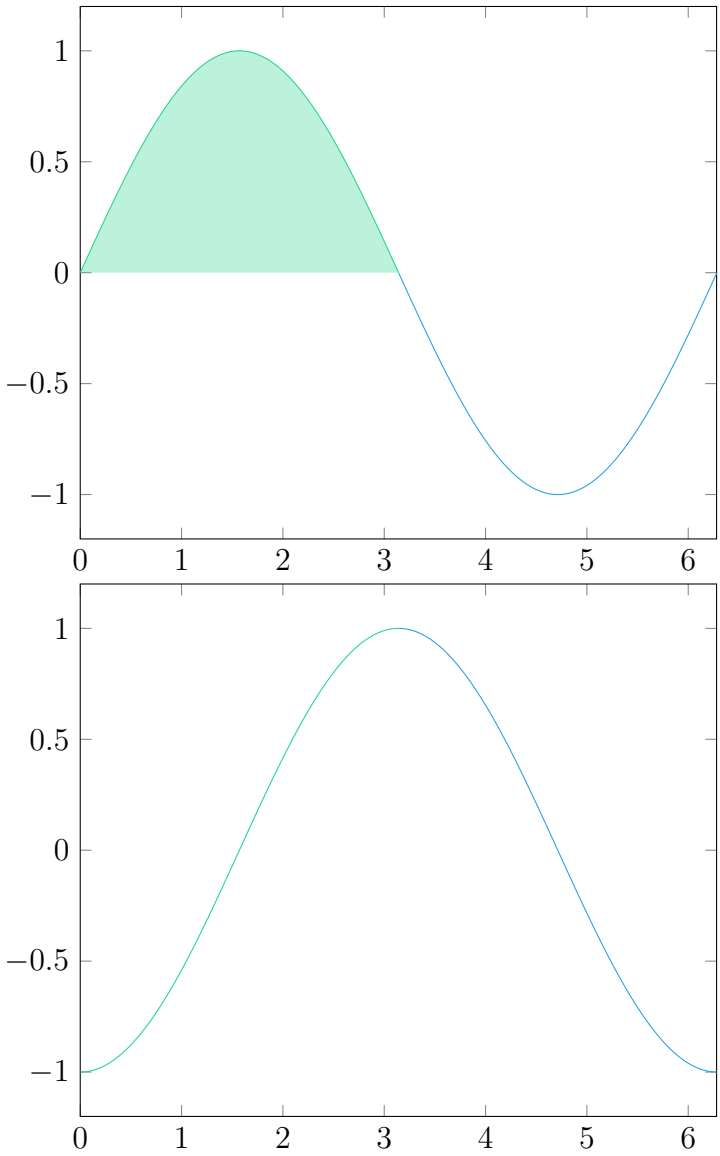


Figure 15: 0¡x¡*pi* range of sin(x) and its integral

and graphing it with the correct values we get the following graph

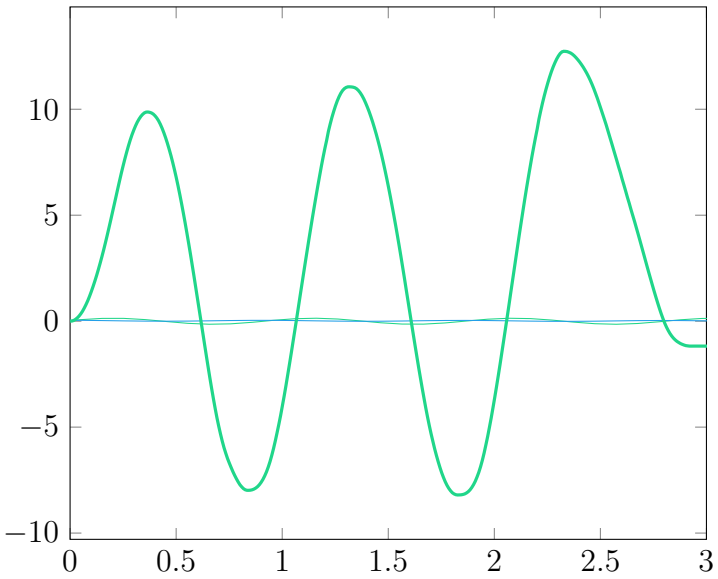


Figure 16: Velocity-time continuous function

Where green is the original function and blue is the integrated function and the green is the actual video recorded displacement. Since the data wasn't scaled accurately, it has led the recorded displacement to be inaccurate.

But this issue is surprisingly similar to the issue we encountered in the numerical approach when we were confused on why the integrated displacement never went down to a negative value.

For now, we will keep in mind that $C_1 = 0.0209628448456$ but continue using C_1 throughout integration for convenience. To find the displacement function with respect to time we integrate again.

5.1 Integrating to displacement

$$s(t) = \int \left(-\frac{a}{b}\cos(bt+c) - dt + C_1\right) \mathrm{d}t \tag{25}$$

$$s(t) = -\frac{a}{b}\int \cos(bt+c)\,\mathrm{d}t - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{26}$$

$$s(t) = -\frac{a}{b}\int \cos(bt+c)\,\mathrm{d}t - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{27}$$

$$\tag{28}$$

Using the same u-substitution:

$$s(t) = -\frac{a}{b}\int \cos(u)\frac{\mathrm{d}u}{b} - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{29}$$

$$s(t) = -\frac{a}{b^2}\sin(u) - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t + C_2 \tag{30}$$

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t + C_2 \tag{31}$$

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \frac{dt^2}{2} + C_1 t + C_2 \tag{32}$$

$$\tag{33}$$

and since this data is filmed as relative displacement from original point (seen as 0) we need to find the value of C_2 such that $s(0) = 0$.

$$s(0) = -\frac{a}{b^2}\sin(b\,(0)+c) - \frac{d\,(0)^2}{2} + C_1\,(0) + C_2 = 0 \tag{34}$$

$$0 = -\frac{a}{b^2}\sin(b\,(0)+c) + C_2 \tag{35}$$

$$\frac{a}{b^2}\sin(b\,(0)+c) = C_2 = 0.000695067592524 \tag{36}$$

$$\tag{37}$$

Again we will keep $C_2 = 0.000695067592524$ in mind. Resulting in a final function $s(t)$:

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \frac{dt^2}{2} + C_1 t + C_2 \tag{38}$$

now if we graph it we get Figure 17

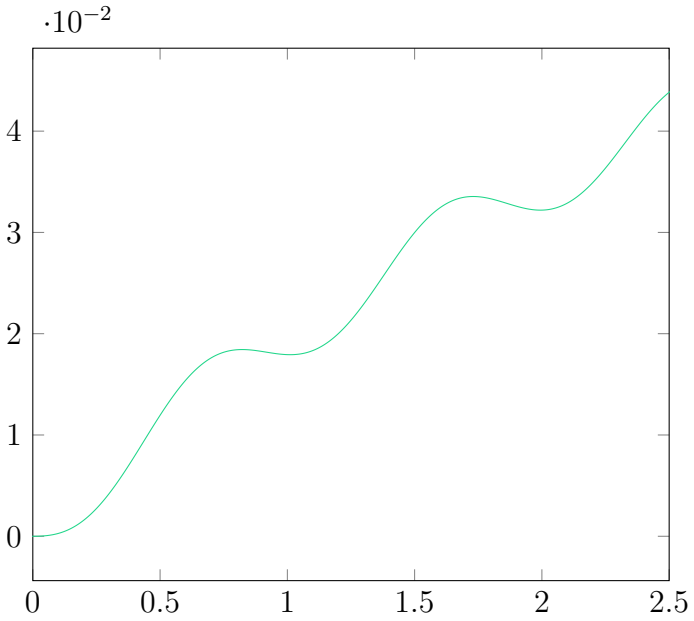


Figure 17: graph of the model being integrated twice

6 Conclusion

We used two methods that both tries to find the displacement-time graph using a given accelerometer-time data, numerically and analytically. For the numerical approach, we have a lot of points that make up a graph when you connect the points. And since there are points, we can use geometry (although it is a lot of geometry) to find the area under the curve made by said points. We centered the data by subtracting each point by the mean of every point's y-value before integrating it numerically to reduce drift during integrated graphs. For the Analytical approach, we fitted a sinusoidal regression on the same accelerometer-time data and then we used algebra and integral calculus to integrate twice to find the displacement-time graph.

6.1 Comparison

Here is the graph before integration for both numerical (the raw data) and the analytical (modeled function from desmos) approach:

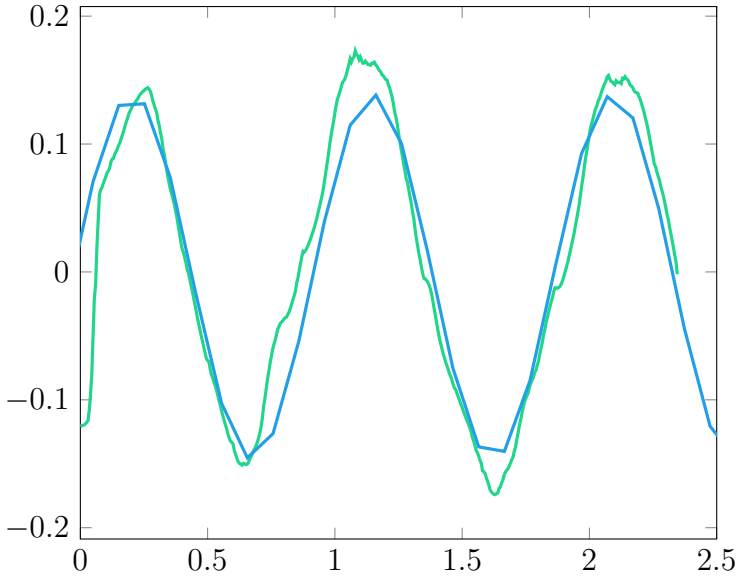


Figure 18: Pre-Integration Graph

The green is the raw data while the blue is the analytical approach. They both start really similar, desmos gives a strong coeeficient of determination of 0.8824, which determines how well the model fits the data.

Here is the both of them in one graph. The green is the numerical method while the blue one is the analytical method and the actual values from the video recording is in orange. We see that for some reason the numerically integrated graph is much more vertically compressed when compared to the analytic function. This is like mentioned before, an issue with the scaling of the actual video recorded data.

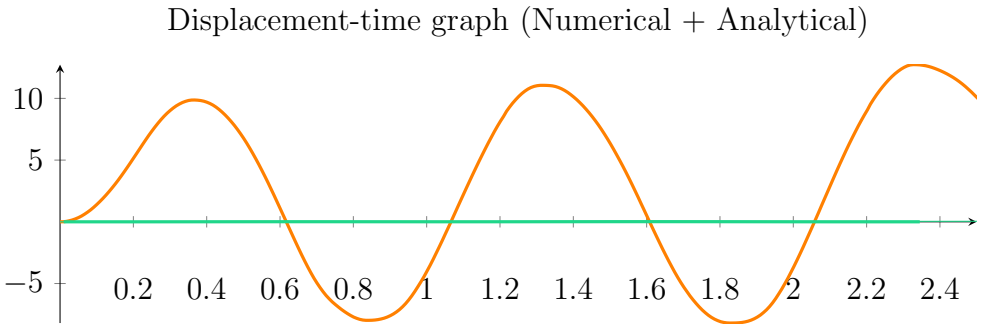


Figure 19: Comparison between integration methods

6.2 Evaluation

While the numerical method follows every accelerometer data point, the analytical method first tries to model it, which makes it a continuous smooth curve and then integrates analytically.

In both cases, a tiny imperfection can get amplified, resulting in drift, but in different ways. In the numerical way, the data is collected in real life and have sensor uncertainties, meaning sometimes it should in theory be one accelerometer value that would result in an integration such that the velocity can start at zero and end at zero. But instead, it is just a little higher or lower, making the mean acceleration higher or lower, and when the mean acceleration is higher or lower, there is a tiny constant change in velocity over time, which results in drift over time. And if you integrate it again, that tiny change gets ever more prominent as time passes. We corrected this drift by making sure the mean is centered before we integrate each time so that the integrated data will not have drift. But there is also sensor uncertainty, which adds more opportunities for drift over time. While numerical methods accumulate uncertainty over time as a result of sensor accuracy. Modeling data with a sinusoidal rergression still can drift, but due to a more mathematical reason. For the equation we decided to model, we already established the fact that the constant d in $a\sin(bx+c)+d$ refers to the mean in terms of its modeled data, and when you integrate any constant, it increases its degree, and making it more impactful over time. However, a more interesting case is when we encounter the first constant C_1 after integration, which turns out to be determined by the phase of our sinusoidal model c. Which intuitively doesn't make sense; why would the phase of an oscillatory function determine the magnitude of drift of its integrated velocity function? Well, its because of our condition that our our initial velocity is 0, but the part in the modeled sine function in which we started recording the data when integrated doesn't have its initial velocity equal to 0, so we have to move it either up or down to compensate, which adds the drift once we integrate again to find displacement.

6.3 Limitations

One limitation we encountered is that the displacement-time graph we got from the numerical method was surprisingly small, it appears to be scaled down by a factor of 1000, although, when you zoom in, it does make sense.

I think this is probably something wrong with the algorithmic issue with my code but it also could be an issue with an erroneous assumption made while making the numerical integration python script. Though, it is linked in the appendix.

Here we see that each method

7 Appendix

7.1 Python code

```
import csv
import math

with open("aligned_best_vel.csv") as dataobject:
    data = tuple(tuple(float(item) for item in row) for row in list(csv.reader(dat
    '''((0.0, -0.120582856), (0.004999756, -0.120273575)'''

    mean = sum(datum[1] for datum in data)/len(data)
    data = [(datum[0], datum[1]-mean) for datum in data]

    vel = []
    accum_vel = 0

    for index,row in enumerate(data):
        try:
            dt = data[index+1][0] - row[0]
            da = data[index+1][1] - row[1]
            dv = dt*row[1] + (1/2) * da*dt

            accum_vel += dv
            vel.append((data[index+1][0], accum_vel))
        except IndexError:
            break

with open('aligned_best_displ.csv', 'w') as velfile:
    file = csv.writer(velfile)
    file.writerow(('Time', 'x'))
    file.writerows(
```

```
[format(datum, 'f') for datum in row]
for row in vel # output source
)
```